Hierarchical Learned Risk-Aware Planning for Modeling Human Driving Behavior in Virtual Environments

Team: Nathan Ludlow, Yiwei Lyu

Sponsor: CMU Center for Autonomous Vehicle Research

Advisors: John Dolan

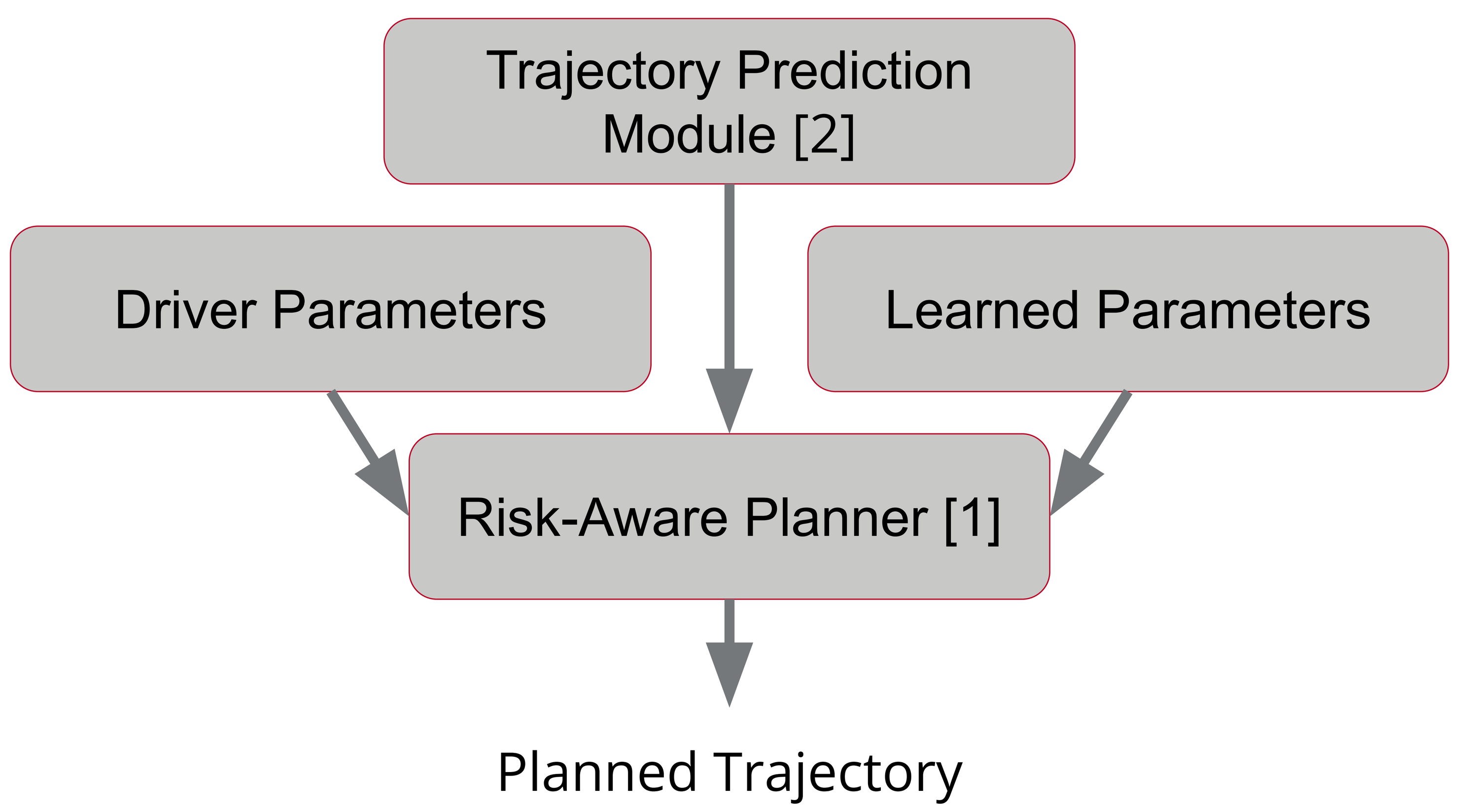
Motivation

As autonomous vehicles continue to permeate modern society, ensuring their safety has emerged as an imperative task. However, the comprehensive real-world testing of these vehicles often proves prohibitively expensive and time-consuming.

Simulation offers an appealing testing alternative due to its cost-effective and high-speed testing capabilities. Yet, to evaluate the safety-critical systems of autonomous vehicles, these must interact with other vehicles. Given that a significant proportion of vehicles are still operated by humans, devising an accurate model of differing human driving behaviors becomes an essential element for thorough testing of autonomous vehicles within simulation environments.

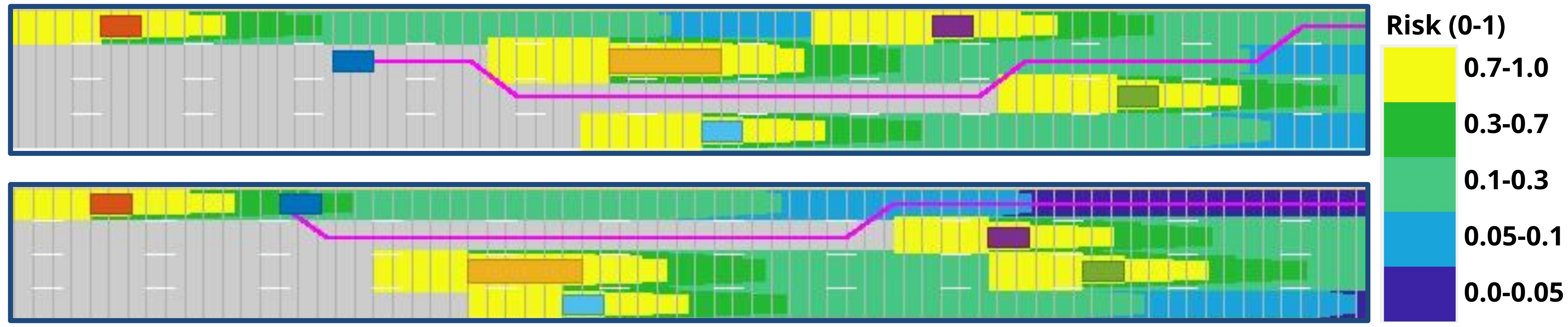
Approach

Our proposed model is a two level hybrid Risk-aware [1], learned model that is inspired by how humans operate their vehicles.



The trajectory prediction module takes vehicle history and predicts potential future trajectories. Each of these trajectories is used to generate risk level fields that represent the potential risk posed to the ego vehicle at a time. The ego vehicle can then plan lowest cost route through the risk. Additional cost factors for path planning are learned from observing human behaviors in data and incorporated into the planner to make the model behave similarly to humans. Driver parameters for the allowable risk are also passed into the model. Then a simple path following controller is implemented to move the vehicle along the trajectory.

Planned Trajectories



This shows two planned trajectories the human driver model will execute. The ego vehicle is shown in blue, and the purple line indicates the planned path the ego vehicle will take while the colored sections of road reflect the risk the driver model perceives at a given location. The first scenario shows an aggressive driver weaving through traffic to travel faster than the other vehicles. The second scenario shows a conservative driver moving out of their lane to allow a faster vehicle to pass them safely.

Learned Parameters

Risk by lane and velocity	Lane 1 (Slowest)	Lane 2	Lane 3	Lane 4	Lane 5 (Fastest)
1 standard deviation below mean velocity	0.4224	0.5030	0.5169	0.4994	0.4460
Mean velocity	0.4017	0.4868	0.4862	0.4677	0.4386
1 standard deviation above mean velocity	0.4123	0.4962	0.5070	0.4891	0.4421

Results

To validate the model we took lane changes found in NGSIM data and replaced the ego car with our proposed model. We recorded the number of lane changes it correctly predicted, as well as the error in the trajectory.

Heuristic	Aggressive Driver	Average Driver	Conservative Driver
Lane Changes Identified	75.06%	79.32%	82.41%
Average RMSE of trajectory	4.88 m	4.21 m	4.25 m
Rinal RMSE of trajectory	6.97 m	6.45 m	6.71 m

Future Work

- Additional validation of model compared to real human trajectories
- Explore modeling more human traits in the framework
- Additional learned parameters fed into the framework
- Addition of stochastic behavior to model
- Incorporation of learned controllers for maneuvering the vehicle along its path [3]

Acknowledgments

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References

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